

BIOL90042 tutorial 6

Multiple choice questions

These are all worth one hypothetical mark, unless otherwise indicated

Question 1

When should I run code in RStudio's console rather than in an R script or Quarto file?

- A. When the code is very important for my analysis
- B. When the code will take a long time to run
- C. When the code is just to quickly check something, and I don't need to keep a record of it
- D. When the code needs to be shared with others

Solution.

Answer: C The console is a good place to try out code you don't want/need to keep a record of, whereas your R script or Quarto file should contain all of the code needed for your analysis

Question 2

I'm trying to read in a file in R, which is called 'ELISA_data.csv' and located in the 'experiments' folder on my desktop. Which of the following `read_csv()` calls is correct?

- A. `read_csv("~/Desktop/experiments/")`
- B. `read_csv("~/Desktop/experiments/elisa_data.csv")`
- C. `read_csv("~/Desktop/experiments/ELISA_data.csv")`
- D. `read_csv("~/Desktop/experiments/ELISA_data.csv")`

Solution.

Answer: D Option A is just the path, with no file name, B has the wrong capitalisation (remember R and computers in general are case sensitive) and C is missing the quotation marks around the path

Question 3 [2 marks]

Fill in the blanks to complete the following paragraph about good coding practice and organisation:

“When doing data analysis, it is very important to stay organised. We can organise all of our scripts and data using a [1], and ensure that our file names are [2]. While writing the actual code, we can use [3] to describe our thought process, and finally share our code with others via [4].”

1. folder / quarto file / R project /
2. detailed / as short as possible / include spaces
3. variables / functions / comments
4. email / git / GitHub

Solution.

Answers: 1. R project 2. detailed 3. comments (while it's true that good variable names help, comments are the main way to describe our code) 4. GitHub (GitHub hosts git repositories in such a way that you can collaborate or share with others, although it's git that handles the actual version control)

Question 4

What is the difference between the git commands 'add' and 'commit'?

- A. Nothing, git add and git commit do the exact same thing
- B. git add is used to save your files, and git commit is used to download them
- C. git add saves your changes locally, while git commit uploads them to GitHub
- D. git add is used to add changes to git's tracking, while git commit saves your changes to the local repository

Solution.

Answer: D You save your files first, then git add can track them, before committing with git commit. It's git push that uploads them to GitHub, or another remote repository.

Question 5

What is quarto?

- A. A way we can combine code and text easily
- B. The same thing as an R script
- C. A new program for running R code
- D. A way to keep track of changes to our code

Solution.

Answer: A Quarto is different to an R script, as we can write text without it needing to be in a comment. Although we can run R code chunks in a quarto document, quarto itself is not running them

Question 6

A scatterplot would **not** be an appropriate visualisation for which pair of variables?

- A. Mouse tail length and whisker length
- B. Survival time and cancer stage (stage 1, 2, 3 and 4)
- C. Drug dosage and gene expression
- D. B cell count and T cell count

Solution.

Answer: B Scatterplots need two continuous variables, and cancer stage is a ordinal categorical variable. Maybe a boxplot or violin plot would be better

Question 7

Which of the following is **not** something we should consider when picking colours for a graph?

- A. Whether the color scheme matches the type of data being shown (e.g., sequential for ordered values, qualitative for categories)
- B. That the intensity of the colours are consistent
- C. Making the sure the colours match the default R colour palette
- D. Making sure it is colourblind-safe

Solution.

Answer: C All of A, B and D are important considerations

Question 8

What are facets, in the context of data visualisation?

- A. A colour palette
- B. A special type of plot
- C. When you colour the points by a column in your dataframe
- D. A way to create several 'mini plots' based on a categorical variable

Solution.

Answer: D Facets are basically mini plots, one for each value of a categorical variable (e.g. one for each mouse strain or one for each cancer subtype)

Question 9

Which of the following is **not** a reason we use replicates in RNA-seq experiments?

- A. Because they increase the sequencing depth of each sample
- B. Because they increase statistical power
- C. Because they allow us to capture biological variation
- D. Because they make us more confident in our results

Solution.

Answer: A Replicates don't increase the sequencing depth, but they do increase our power, and make us more confident by capturing more of the biological variation

Question 10 [2 marks]

Fill in the blanks to complete the following paragraph about biological data types:

"There are many different technologies that can be used to generate biological data. For example, if we are interested in knowing the sequence of DNA, we can use [1]. Many types of biological data require specialised analysis techniques, like [2] for transcriptomic and proteomic data. After this, we are generally interested in finding things that are [3], which we can visualise on a [4]."

1. NGS / mass spectrometry / microscopy
2. squaring / log-transformation / pivoting
3. differentially expressed / very high / very low
4. bar chart / boxplot / volcano plot

Solution.

Answers: 1. NGS 2. log-transformation (which is used to transform highly skewed data to become normally distributed) 3. differentially expressed (this includes both things that are higher and lower than usual) 4. volcano plot

Question 11

Which of the following is an advantage of using multiple linear regression over simple linear regression?

- A. Because it removes the need to check any model assumptions
- B. Because it guarantees the relationship between variables is causal
- C. Because it allows us to examine the effect of multiple variables on an outcome
- D. Because it makes our correlation higher

Solution.

Answer: C Multiple linear regression still has the same assumptions as simple linear regression, and it doesn't affect the correlation beyond possibly fitting the data better but this is not guaranteed

Question 12

Why is it important to correct for batch effects in RNA-seq?

- A. Because batch effects always increase the total number of genes detected
- B. Because batch effects can be mistaken for real biological differences, leading to false conclusions
- C. Because correcting for batch effects removes the need for biological replicates
- D. Because we want all of our data to be as similar as possible

Solution.

Answer: B Batch effects can confound our interpretation of the biology, which is why we remove them but they come from technical variation so biological replicates are still required

Question 13

Which of the following is **not** a way we can assess the validity of our linear regression model?

- A. By looking at p-values for the model coefficients
- B. By checking residual plots for patterns (to assess linearity and homoscedasticity)

- C. By examining the R^2 value to see how much variance is explained
- D. By checking whether there is a positive correlation between the dependent and independent variables

Solution.

Answer: D The correlation could be negative as well! The rest are all useful things to look at.

Question 14

What is the atomic type of the value "TRUE" ?

- A. Data frame
- B. Logical (boolean)
- C. Character
- D. Numeric

Solution.

Answer: C Because of the quotation marks, it becomes a character, even though TRUE on its own would be logical

Question 15

What would be the result of running the following code: `summary(iris$Sepal.Length)`

- A. An error
- B. Various statistics about the `iris` dataframe
- C. The mean of the `Sepal.Length` column in the `iris` dataframe
- D. Various statistics about the `Sepal.Length` column in the `iris` dataframe

Solution.

Answer: D The `$` allows us to access the `Sepal.Length` column, and `summary()` calculates various summary statistics, including but not limited to the mean.

Question 16

Which of the following would sort the `temp` column of `beaver1` in descending order?

- A. `beaver1 %>% arrange(temp)`
- B. `beaver1 %>% arrange(desc(temp))`
- C. `beaver1 %>% sort(temp)`

D. `beaver1 %>% slice_max(temp, n = 5)`

Solution.

Answer: B The sort function in the tidyverse is called `arrange()` and we need the helper function `desc()` to sort in descending order.

Question 17

I'm trying to add a new column to `beaver1` that contains the temperature in Farenheit. I know the formula for this is $^{\circ}F = (^{\circ}C \times 9/5) + 32$, so I have written the following code:

```
beaver1 %>% mutate(temp_f == (temp * 9/5) + 32)
```

However, I get an error: `! object 'temp_f' not found`

What is the problem?

- A. There is a missing bracket
- B. `temp_f` doesn't exist as a column, it should just be `temp`
- C. The use of `==` inside the `mutate()` function
- D. You can't use the `mutate()` function for this

Solution.

Answer: C The problem is the `==`, which is only used for logical tests, which this is not. It should just be a single `=`. It is true that `temp_f` doesn't exist in `beaver1` as a column, but that is what the `mutate()` function is trying to do here, to create that column

Question 18

Which of the following would **not** allow me to join two dataframes `df1` and `df2`, by a shared column `col1`?

- A. `left_join(df1, df2, col1)`
- B. `left_join(df1, df2)`
- C. `df1 %>% left_join(df2)`
- D. `left_join(df1, df2, by = "col1")`

Solution.

Answer: A To join by col1, it must be supplied as a character string using the `by =` argument, not just along with the dataframes. The other options are all correct, if you don't specify which column to join by as in option D, `left_join()` will join automatically by all shared columns (options B and C)

Question 19

In the `pivot_longer()` function, the `names_to` argument refers to:

- A. The names of columns to pivot
- B. The name of a new column which will contains the pivoted values
- C. The name of the dataframe that you want to pivot into
- D. The name of a new column which will contain the names of the columns you pivoted

Solution.

Answer: D `names_to` is the column containing the names of the columns that have been pivoted longer. Option A is the `cols` argument and option B is the `values_to` argument

Question 20

In the `ggplot2` package, what is the `aes()` function used for?

- A. To assign columns from a dataframe to various aesthetics of a plot, like the x and y axes, or colour
- B. To add points to a plot
- C. To create the initial setup of the plot
- D. To describe purely aesthetic features of the plot, like the font size or whether something should be bold

Solution.

Answer: A The `aes()` function is used to assign columns to aesthetics of the plot like x, y, colour, fill, shape etc. Adding points to a plot is done with `geom_point()`, initial setup of the plot is done with `ggplot()` and you set font size etc in other functions, e.g. `theme()`

Question 21

I'm trying to make a boxplot, of the `temp` column in `beaver1`, but I got an error. This is my code:

```
beaver1 %>% ggplot(x = temp) + geom_boxplot()
```

And this is the error message: `! stat_boxplot() requires an x or y aesthetic.`

What is wrong?

- A. The function is called `ggplot2()` not `ggplot()`
- B. `x = temp` should be inside the `aes()` function
- C. You can't use the pipe `%>%` with `ggplot2`
- D. To make a boxplot you should use the `boxplot()` function, not `geom_boxplot()`

Solution.

Answer: B Because the `aes()` function is missing, `ggplot2` thinks that you haven't given it an x or y aesthetic (data) to plot

Question 22

Describe in words what the following code does:

```
airquality %>%  
  replace_na(list(Ozone = 0)) %>%  
  ggplot(aes(x = Ozone, y = Temp)) +  
    geom_point()
```

- A. Nothing, this code would give an error
- B. Removes NA in the ozone column, then makes a scatterplot
- C. Creates a scatterplot of ozone and temperature, after replacing missing values in the ozone column with 0
- D. Replaces all missing values with 0 and then makes a scatterplot of ozone and temperature

Solution.

Answer: C Note that `replace_na()` only replaces (not removes) missing values in the columns specified inside the `list()`, in this case ozone

Question 23

Why is it important for data to be tidy?

- A. Because it makes our code run faster
- B. Because it allows us to consistently re-use a set of functions across many datasets, which are known as the tidyverse
- C. Because it removes outliers and makes our analysis look cleaner
- D. Because tidyverse functions only work on tidy data

Solution.

Answer: B Note that option D is incorrect as most functions in tidyverse only work on tidy data, but there are also many functions that exist specifically to clean/wrangle the data (e.g. the pivot functions)

Short-answer questions

Question 1 [2 marks]

I've written the following R code:

```
rnaseq_data %>%
  summarise(
    min_expression = min(expression),
    .by = c(group, gene)
  )
```

Explain in words what it does. Can you think of another function that could be used to do the same thing?

Solution.

Answer: This code finds, for each unique combination of 'group' and 'gene', the minimum expression. This is what the `slice_min()` function does internally, it's just wrapped up in a convenient helper function for you.

Question 2 [4 marks]

I've written the following R code, but it contains some errors

```
dosage %>%
  filter(mouse_strain = "CD-1") %>%
  select(id_num, weight_lost_g, cage_number)
  summarise(
    mean_weight_lost = mean(weight_lost_g),
    by = cage_number %>%
  pivot_wider(
    names_from = cage_number,
    values_to = mean_weight_lost)
```

```
Error in parse(text = input): <text>:11:0: unexpected end of input
9:     values_to = mean_weight_lost)
10:
   ^
```

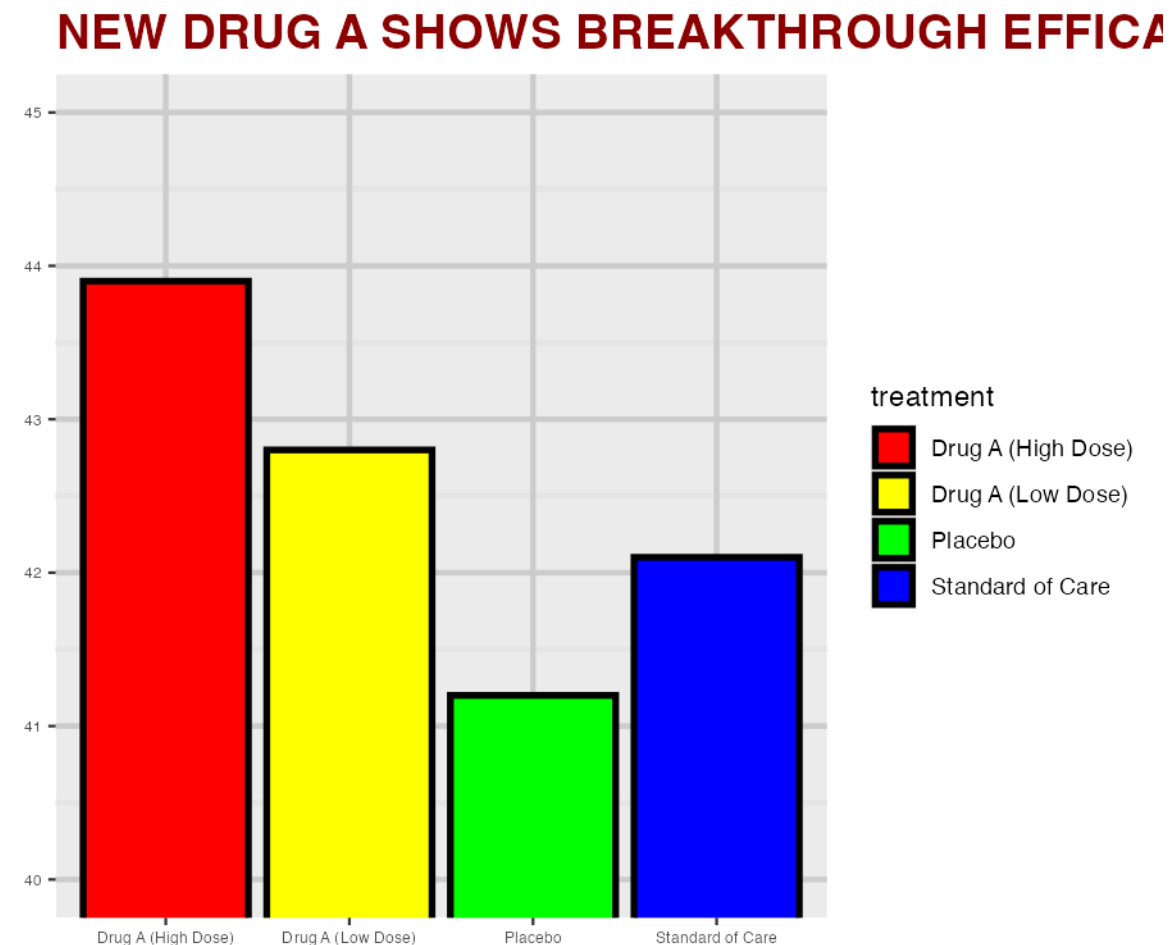
Identify **four** errors in the code (there may be more)

Solution.

Answer: There are many errors in this code. - in the `filter()` function, it should be `==` not `=` - after the `select()` function, the pipe is missing - in the `summarise()` function, the argument for grouping is `.by` not `by` - the `summarise()` function is also missing a bracket - in the `pivot_wider()` function, the argument is `values_from` not `values_to`

Question 3 [4 marks]

My colleague has made this interesting graph:



Do you think this plot is easy to interpret? Justify your answer by identifying **three** elements of the plot that violate good data visualisation principles.

Solution.

Answer: No, this plot is so bad!

- the title is cut off and so can't be read
- the colour palette is really distracting and makes it painful to look at
- the axes ticks are so small it's hard to read
- the y axis doesn't start at zero, so the relatively small differences (~41 for placebo vs ~44 for drug A) are exaggerated
- the y axis has no label, what do the numbers refer to?